

TECHNICAL ASSESSMENT

Product Recommendation Similarity System : Design Brief

For the role of Machine Learning / NLP Engineer (Search & Recommendations)

ISSUED BY	Acme Retail, Data Science & Personalisation Group
ROLE	Machine Learning / NLP Engineer (Search & Recommendations)
ASSESSMENT TYPE	Take-home design and implementation exercise
DATE OF ISSUE	30 June 2026
TIME ALLOWED	48 hours from receipt

1. CONTEXT

The candidate is being evaluated for a Machine Learning and NLP role at Acme Retail. Here, the engineering teams build production-ready search, discovery, and personalisation systems for a high-volume e-commerce marketplace. Product discovery means helping a customer move from one product to other related products. This is very important for the customer experience and also for business growth.

This assessment gives you a problem that is similar to the actual work done by the team. The aim is to check your ability to design, build, and clearly explain a working solution. The solution should understand the meaning-based relationship between products, instead of depending only on exact keyword matching.

1.1 About the Role (NLP / ML Engineer)

An NLP (Natural Language Processing) Engineer works with text data and builds systems that can understand human language. In a search and recommendations team, this role usually involves:

- (a) cleaning and preparing text such as product titles, descriptions, and reviews;
- (b) converting text into numerical vectors (embeddings) that capture meaning;
- (c) building similarity, ranking, and retrieval logic for search and recommendations; and
- (d) testing the quality of the results and improving them over time.

A good NLP / ML Engineer should be comfortable on both sides: the language side (text processing) and the engineering side (building reliable, production-ready systems).

2. PROBLEM STATEMENT

Design and build a system that takes one product as input and returns the most similar products from a catalogue. Here, similarity should be based on the actual meaning of each product's text fields (such as title, description, and category), and not only on simple word matching.

3. OBJECTIVE

For a given query product, which can be either a catalogue ID or plain product text, the system should:

- (a) represent each product in a form that captures its meaning;
- (b) compare the query product with the rest of the catalogue using a proper similarity measure; and

- (c) return a ranked list of the top-N most similar products, arranged by relevance,

so that the recommendations are clearly related to the query product and would interest a customer who is viewing it.

4. FUNCTIONAL REQUIREMENTS

4.1 Text Attribute Support

The system should accept and process the text fields of a product. At a minimum, it should use the product title and description. Wherever available, it can also use category, brand, or tag fields. It should work well for both short titles and long, free-text descriptions.

4.2 Representation and Embeddings

The system should convert product text into vector representations that capture meaning. Candidates are expected to go beyond a simple word-matching baseline. Dense embedding methods (for example Word2Vec, FastText, or other word or sentence embedding methods) are encouraged. A simple baseline such as TF-IDF can be added for comparison, but you should understand and explain the limits of using it alone.

4.3 Similarity and Ranking

The system should calculate a similarity score between products using a suitable measure (for example, cosine similarity on the chosen representation). It should then return the top-N most similar products for any queried item, arranged from highest to lowest similarity.

4.4 Processing Pipeline

The system should take a product catalogue and a query product as input. It should do all the required processing inside the system itself, such as text cleaning, representation, similarity calculation, and ranking. Finally, it should return the recommendations without needing any extra input from the user.

4.5 Generalisation

The system should not be fixed or limited to only one catalogue. It should give sensible recommendations even for new products that were not seen before. When it comes across new words that were not seen during development, it should still work in a reasonable way without breaking.

5. INPUT AND OUTPUT SPECIFICATION

Input	A product catalogue in a common structured format (for example, CSV or JSON). It should contain at least a product ID and one or more text fields such as title and description. Along with this, a query product is given, either as a catalogue ID or as plain product text.
Output	A ranked list of the top-N most similar products to the query. Each product should come with its similarity score. The output should be in a standard structured form (for example, a printed table, CSV, or JSON).

6. EVALUATION CRITERIA

Submissions will be checked using the following criteria. Candidates are advised to clearly address each point in their submission.

Criterion	Description
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6.1 Recommendation Quality	How relevant and useful the returned products are for a query. This will be judged in a qualitative way and, where suitable, using standard quantitative checks.
6.2 Use of Representations	How suitable and correct the chosen representation is. In particular, the use of meaning-based embeddings instead of a keyword-only baseline, and how sound the similarity logic is.
6.3 Robustness	How well the system handles products with short or long text, missing or limited fields, and different product categories.
6.4 Generalisation	How well the system performs on products and words that were not seen during development or training.
6.5 Clarity of Approach	The quality, structure, and correctness of the candidate's explanation of their method.

7. SUBMISSION REQUIREMENTS

Along with the working system, candidates must also submit a short explanation video. The video is a compulsory part of this assessment. At a minimum, it should cover:

- (a) your overall approach to the problem and the reason for choosing your representation and similarity technique;
- (b) a clear explanation of how the system works inside, including text pre-processing, the embedding or representation stage, and the similarity and ranking stage; and
- (c) a step-by-step dry run on at least one example query product, showing how the input changes at each stage of the pipeline and how the final ranked recommendations are produced.

Submissions without the explanation video will be treated as incomplete.

8. TIMELINE

The assessment should be completed and submitted within **forty-eight (48) hours** from the time of issue. Late submissions will not be accepted unless an extension has been agreed in writing in advance.

Note to candidate: You are free to use any publicly available libraries, frameworks, datasets, or pre-trained models, as long as you clearly mention all such usage in your explanation. Original thinking is valued, and so are good engineering judgement and clear communication. If anything in this document is not clear, please ask for clarification before you start the work.